Problem Statement

Evaluate the limit from the image "dailyintegral-limit-limit-1x1-2026-07-27.jpg":

Let $a_1 = 1$ and $a_n = n(a_{n-1} + 1)$ for $n \geq 2$. Define the product $P_n = \prod_{k=1}^{n} \left(1 + \frac{1}{a_k}\right)$. Find:

$$\lim_{n\to\infty} P_n$$

Detailed Solution

To find the limit of the infinite product, we first compute the initial terms of the sequence a_n and the partial products P_n to identify a pattern. Then, we prove a closed-form formula for P_n using mathematical induction, solve the underlying linear recurrence relation, and evaluate the final series limit.

Step 1: Pattern Recognition and Inductive Proof

We begin by generating the first few terms of the sequence a_n using the recurrence relation $a_n = n(a_{n-1} + 1)$ with $a_1 = 1$:

$$\begin{aligned}
a_1 &= 1 \\
a_2 &= 2(1 + 1) = 4 \\
a_3 &= 3(4 + 1) = 15 \\
a_4 &= 4(15 + 1) = 64
\end{aligned}$$

Next, we evaluate the first few partial products $P_n = \prod_{k=1}^{n} \left(1 + \frac{1}{a_k}\right)$:

$$\begin{aligned}
P_1 &= 1 + \frac{1}{1} = 2 = \frac{2}{1!} \\
P_2 &= P_1 \cdot \left(1 + \frac{1}{4}\right) = 2 \cdot \frac{5}{4} = \frac{5}{2} = \frac{5}{2!} \\
P_3 &= P_2 \cdot \left(1 + \frac{1}{15}\right) = \frac{5}{2} \cdot \frac{16}{15} = \frac{8}{3} = \frac{16}{3!} \\
P_4 &= P_3 \cdot \left(1 + \frac{1}{64}\right) = \frac{8}{3} \cdot \frac{65}{64} = \frac{65}{24} = \frac{65}{4!}
\end{aligned}$$

Notice that each partial product can be expressed neatly in terms of the sequence term a_n and the factorial of the index n. We claim that for all $n \geq 1$:

$$P_n = \frac{a_n + 1}{n!}$$

Proof by Mathematical Induction:

- **Base Case ($n = 1$):** We have $P_1 = 2$ and $\frac{a_1+1}{1!} = \frac{1+1}{1} = 2$. The formula holds for $n = 1$.

- **Inductive Step:** Assume the formula holds for some integer $k \geq 1$, meaning $P_k = \frac{a_k+1}{k!}$. For $n = k+1$, the partial product is given by:

$$P_{k+1} = P_k \cdot \left(1 + \frac{1}{a_{k+1}}\right) = \frac{a_k + 1}{k!} \cdot \frac{a_{k+1} + 1}{a_{k+1}}$$

From the given recurrence relation, we know that $a_{k+1} = (k+1)(a_k+1)$, which implies $a_k + 1 = \frac{a_{k+1}}{k+1}$. Substituting this into our expression yields:

$$P_{k+1} = \frac{\frac{a_{k+1}}{k+1}}{k!} \cdot \frac{a_{k+1} + 1}{a_{k+1}} = \frac{a_{k+1}}{(k+1)!} \cdot \frac{a_{k+1} + 1}{a_{k+1}} = \frac{a_{k+1} + 1}{(k+1)!}$$

This completes the induction.

Step 2: Solving the Recurrence Relation for a_n

To evaluate $\lim_{n\to\infty} P_n$, we need to determine the asymptotic behavior of $\frac{a_n}{n!}$. Let $b_n = \frac{a_n}{n!}$. Dividing both sides of the original recurrence relation $a_n = n(a_{n-1} + 1)$ by $n!$, we obtain:

$$\frac{a_n}{n!} = \frac{n(a_{n-1} + 1)}{n!} = \frac{a_{n-1} + 1}{(n-1)!} = \frac{a_{n-1}}{(n-1)!} + \frac{1}{(n-1)!}$$

In terms of our auxiliary sequence b_n, this simplifies to a linear difference equation:

$$b_n = b_{n-1} + \frac{1}{(n-1)!} \quad \text{for } n \geq 2$$

Given the initial condition $b_1 = \frac{a_1}{1!} = 1 = \frac{1}{0!}$, we can expand this relation by telescoping:

$$b_n = b_1 + \sum_{k=1}^{n-1} \frac{1}{k!} = \frac{1}{0!} + \sum_{k=1}^{n-1} \frac{1}{k!} = \sum_{k=0}^{n-1} \frac{1}{k!}$$

Step 3: Evaluating the Infinite Product Limit

Now, we substitute our closed-form expression for $b_n = \frac{a_n}{n!}$ back into the formula for the partial product P_n:

$$P_n = \frac{a_n + 1}{n!} = \frac{a_n}{n!} + \frac{1}{n!} = b_n + \frac{1}{n!} = \sum_{k=0}^{n-1} \frac{1}{k!} + \frac{1}{n!} = \sum_{k=0}^{n} \frac{1}{k!}$$

Thus, the partial product P_n is precisely the n-th partial sum of the standard Taylor series expansion for the exponential function evaluated at $x = 1$. Taking the limit as $n \to \infty$:

$$\lim_{n\to\infty} P_n = \lim_{n\to\infty} \sum_{k=0}^{n} \frac{1}{k!} = \sum_{k=0}^{\infty} \frac{1}{k!} = e$$

Final Answer:

$$\mathbf{e}$$